

FCT SERVICE NOTE SN05091210f

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Date: September 30, 2008

Subject: VCT Detector Center Module Failures & Local FE Support actions.

Effectivity: VCT 32/64 slice systems (software releases up to and including Service Pack 2)

Details:

There are two types of VCT Das/Detectors released, VDAS (Sherlock Detector) and HDAS (HALO Detector). They are identified on the Common Service Desktop DAS type as VDAS and HDAS.

Configuration file edits are necessary when performing a center module replacement (slots 27-31, center five). This service note defines the actions necessary during center module replacement. Service Pack 3 is expected to obsolete all manual file modifications defined in this service note.

For **VDAS64/HDAS64** The five Center Module FRUs (slots 27-31) are a different part number from outer module FRU. The part number for a VDAS64 center module is 5136400-3. The part number for the HDAS64 center module is 5196401. For **VDAS32** the part number of the detector module is common between center and outer modules (P/N 5134959).

Resolution:

See the following pages for the 32 and 64 slice actions necessary when replacing a VCT center module.

For VDAS64, HDAS64 and VDAS32:

Perform the following steps based on your software release that is either 07MW18.4 or Service Pack 2 loaded on 07MW18.4.

Note: For VDAS32, replace “FRDMTool64” with “FRDMTool32”.

For 07MW18.4:

1) Prior to replacing the module you must perform the following steps in a Unix shell:

```
{ctuser}[1] cd /usr/g/config
```

```
{ctuser}[2] cp FRDMTool64.cfg FRDMTool64.cfg.lfc (makes a backup copy)
```

```
{ctuser}[3] nedit FRDMTool64.cfg
```

Edit the following lines (do NOT edit any other config lines):

ENABLE_BIS_CHECKS = ON (change from OFF to ON if not already ON)

BIS_MATLAB_TEST_THRESHOLD_ARGV = 1.5e-6 (change to 1.5 if not already set to 1.5)

Save the change and exit the editor.

2) Start the FRDM Wizard and follow the module replacement process.

3) When done with the replacement process turn off the BIS_CHECKS (center module test) in the config file. This can be done while calibrations are running.

```
{ctuser}[1] nedit FRDMTool64.cfg
```

ENABLE_BIS_CHECKS = OFF (turns off the center module test for any other outer module replacements performed later)

For 07MW18.4 with Service Pack 2:

1) Prior to replacing the module you must perform the following steps in a Unix shell if not already done previously, only need to perform again if a software reload has occurred:

```
{ctuser}[1] cd /usr/g/config
```

```
{ctuser}[2] cp FRDMTool64.cfg FRDMTool64.cfg.lfc (makes a backup copy)
```

```
{ctuser}[3] nedit FRDMTool64.cfg
```

Edit the following lines (do NOT edit any other config lines):

BIS_MATLAB_TEST_THRESHOLD_ARGV = 1.5e-6 (change to 1.5 if not already set to 1.5)

Save the change and exit the editor.

2) Start the FRDM Wizard and follow the module replacement process.

Service Pack 2 automated the functionality of only running the center module test (BIS_Check) when the FRDM wizard sees a center module change.

If the FRDM wizard indicates a center module test failure (run right after selecting “Prep for Calibrations” button) then you have two potential recovery methods.

A failure indicates that the module just put in is not compatible with the existing modules and may create a center artifact in some scan mode.

- 1.) Finish calibrations and IQ checks and let the customer run while ordering a new center module to replace at another scheduled time.

NOTE: module needs to be replaced as the failure indicates a center artifact could occur.

- 2.) Run the DDC scans indicated in Appendix A section at the bottom of this service note and follow the instructions there, calling OLC to coordinate a module swap plan with engineering. Engineering will respond with a plan that will swap the failing module with another module in the central region of the detector to alleviate the problem if possible per supplied data. The “createMasterDCF” script must be run in a unix shell after a special/Engineering module swap to allow the FRDM wizard to continue.

Appendix A:

DDC Scans (64 slice systems)

- 1) Run the following 2 DDC scans. This data will be used to create the backup module plan by the detector engineering team.
 - Scan 1: kv140, 100mA stationary 1 second large focal spot, 64x0.625 scan air filter
 - Scan 2: kv80, 462mA stationary 1 second large focal spot, 64x0.625 scan air filter
- 2) Using Scan Analysis utility, create MSD dd files with offset correction and reference normalization selected for preprocessing.
- 3) Record dd file names
- 4) Select DD Tab at top of scan analysis – click refresh select dd file just created.
- 5) Select MSD dd files and Save as ASCII
- 6) Record ASCII file name. (example: dd.VCT1.50009.2.1.000000.msd.txt)

Ask OLC to pull files back for engineering evaluation and ask them to contact detector engineering per the center module support process they have been given. Files are found in the /usr/g/service/dd directory.

DDC Scans (32 slice systems)

- 1) Run the following 4 DDC scans. This data will be used to create the backup module plan by the detector engineering team.
 - Scan 1: kv140, 100mA stationary 1 second large focal spot, 32x0.625 scan air filter
 - Scan 2: kv80, 462mA stationary 1 second large focal spot, 32x0.625 scan air filter
 - Scan 3: kv140, 100mA stationary 1 second large focal spot, 32x1.25 scan air filter
 - Scan 4: kv80, 462mA stationary 1 second large focal spot, 32x1.25 scan air filter
- 2) Using Scan Analysis utility, create MSD dd files with offset correction and reference normalization selected for preprocessing.
- 3) Record dd file names
- 4) Select DD Tab at top of scan analysis – click refresh select dd file just created.
- 5) Select MSD dd files and Save as ASCII
- 6) Record ASCII file name. (example: dd.VCT1.50009.2.1.000000.msd.txt)

Ask OLC to pull files back for engineering evaluation and ask them to contact detector engineering per the center module support process they have been given. Files are found in /usr/g/service/dd directory.

Updates to this service note will be issued when the above steps are no longer required or any part of the VCT center module process changes.

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